

## **Experiment 14:**

### **Implement House price Prediction using appropriate machine learning algorithm**

#### **Python Code:**

```
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([
    [1000, 2, 10],
    [1200, 2, 8],
    [1500, 3, 7],
    [1800, 3, 5],
    [2000, 3, 4],
    [2200, 4, 3],
    [2500, 4, 2],
    [2800, 4, 1],
    [3000, 5, 1],
    [3500, 5, 0]
])

y = np.array([
    3000000, 3500000, 4500000, 5500000, 6000000,
    6800000, 7500000, 8500000, 9500000, 11000000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

new_house = np.array([[2000, 3, 4]])
```

```
predicted_price = model.predict(new_house)
print("Actual prices:", y_test)
print("Predicted prices:", np.round(y_pred, 2))
print("Mean Squared Error:", round(mean_squared_error(y_test, y_pred), 2))
print("R2 Score:", round(r2_score(y_test, y_pred), 2))
print("New house details:", new_house[0])
print("Predicted house price:", round(predicted_price[0], 2))
```

**Output:**

```
Actual prices: [9500000 3500000]
Predicted prices: [9372241.99 3497953.74]
Mean Squared Error: 8163147788.21
R2 Score: 1.0
New house details: [2000    3    4]
Predicted house price: 5977846.98
```

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